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Phase I Study of the Anti-interleukin 33 Fragment Antigen-Binding Region RO7303359 in Geographic Atrophy Secondary to Age-Related Macular Degeneration

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Purpose: Interleukin (IL) 33 is a potent proinflammatory cytokine and a potential target in age-related macular degeneration (AMD) pathophysiology. This study evaluated RO7303359, an anti—IL-33 fragment antigen-binding region (Fab) targeting the IL-33/serum stimulation-2 (ST2) pathway, in patients with geographic atrophy (GA) secondary to AMD.

Design: Phase I, open-label, multicenter, single-dose, dose-escalation study.

Participants: Patients with GA secondary to AMD.

Methods: Patients received a single intravitreal (IVT) injection of RO7303359 (dose range, 1—20 mg).

Main Outcome Measures: The primary objective was to evaluate the safety and tolerability of RO7303359. The secondary objectives included assessing pharmacokinetics (PK), immune response, and pharmacodynamic (PD) biomarker activity. Pharmacodynamic biomarkers assessed in aqueous humor included CC motif chemokine ligand 2, CXC motif chemokine ligand 10, IL-6, and intercellular adhesion molecule 1.

Results: Thirty-seven patients enrolled in the dose cohorts. Single IVT doses of RO7303359 demonstrated an acceptable safety profile up to 20 mg, with mild ocular adverse events reported. Pharmacokinetics analysis revealed dose-proportional exposure within expected ranges for an IVT-administered Fab. No treatment-emergent antidrug antibodies were observed. Evaluation of changes in aqueous humor PD biomarker levels failed to demonstrate a discernible effect on IL-33/ST2 pathway activity.

Conclusions: In this trial (ClinicalTrials.gov identifier, NCT04615325), while RO7303359 exhibited acceptable safety and PK profiles, absence of demonstrable effects on the IL-33/ST2 pathway using aqueous PD biomarkers resulted in discontinuation of development in GA. Further work is needed to evaluate the relevance of the IL-33/ST2 pathway in the pathophysiology of AMD.

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Supplemental material available at www.ophthalmologyscience.org.

Geographic atrophy (GA) is an advanced stage of age-related macular degeneration (AMD) characterized by degradation of the retinal pigment epithelium, photoreceptors, and choriocapillaris.[1,2] The worldwide incidence of GA is estimated to be >5 million people, with incidence increasing exponentially with age, resulting in a substantial global economic burden.[3] Retinal pigment epithelium and photoreceptor degeneration expand to encompass the macula and the fovea, leading to irreversible loss of vision.[4,5] Two agents have been approved in the United States. Pegcetacoplan and avacincaptad pegol offer reductions in the rate of GA progression when administered as a series of intravitreal (IVT) injections, every 25 to 60 days for pegcetacoplan[6] or once monthly for up to 12 months for avacincaptad pegol.[7] Despite these recent advances, there remains a significant unmet medical need for treatment of GA with therapies offering substantial slowing or even arrest of disease progression and vision loss.

Interleukin (IL) 33 is a tissue-derived cytokine that potently drives proinflammatory signaling. It functions as an alarm signal (alarmin) released upon cell injury or tissue damage to alert the immune cells expressing the serum stimulation-2 (ST2) receptor (IL-1 receptor-like 1).[8] In the human eye, IL-33 is primarily expressed in glial, retinal pigment epithelial, and endothelial cells.[9] In patients with

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AMD, the number of IL-33+ Müller cells is increased in areas of retinal atrophy compared with adjacent unaffected areas and compared with the macula of aged normal controls, and levels of IL-33 were increased in vitreous.[9] In support of a potential role of IL-33 in AMD pathophysiology, introduction of phototoxic retinal stress activates the IL-33/ST2 pathway, leading to proinflammatory cytokine secretion, phagocytic cell recruitment into the outer retina, and photoreceptor degeneration.[9] Additionally, deletion of the ST2 receptor or inhibition of IL-33 reduces the release of proinflammatory mediators, decreases accumulation of phagocytes in the retina, and protects photoreceptors after constant light exposure.[9] These results suggest that IL-33 signaling might play a role in the retinal damage and associated proinflammatory environment in GA.[10] Interleukin 33 possesses both proangiogenic and antiangiogenic roles. This role appears to be dependent on the stimulus, insult context, and tissue type,[11,12] and its relevance is unclear in the context of GA secondary to AMD.

RO7303359 (FITT9122A, anti–IL-33 fragment antigen-binding region [Fab]) is a humanized antigen binding fragment (Fab) that selectively binds to the active (reduced) form of human IL-33 and blocks its binding to its high affinity receptor ST2. Based on the results of the nonclinical pharmacokinetics (PK) and toxicology studies, RO7303359 may represent a novel therapeutic option for the treatment of GA by blocking the activation of the IL-33/ST2 pathway and decreasing inflammation, which may slow down the progression of the retinal degeneration in patients with GA secondary to AMD.

Methods

Ethics Statement

The study protocol was approved by Institutional Review Boards before patient recruitment and conducted in accordance with the International Conference on Harmonization E6 Guidelines for Good Clinical Practice. Each subject provided written informed consent for participation in the study before enrollment. ClinicalTrials.gov Identifier: NCT04615325. This study adhered to the tenets of the Declaration of Helsinki.

Study Objectives

This was an open-label, multicenter, single-dose, dose-escalation, phase Ia study. The primary objective was to investigate the ocular and systemic safety and tolerability of RO7303359 after a single IVT dose and to determine the maximum tolerated dose (MTD). The secondary and exploratory objectives included characterization of the PK properties, evaluation of immune response, and assessment of pharmacodynamic (PD) biomarker activity of RO7303359. Patients were enrolled and followed at 11 sites in the United States between December 8, 2020 and January 4, 2023.

Study Population

Patient-level inclusion criteria included age ≥50 years. The key ocular inclusion criteria of the study eye included a well-demarcated area of GA secondary to AMD in the absence of choroidal neovascularization, with a GA lesion area ≥0.5 disc area (1.25 mm^2) and a best-corrected visual acuity (BCVA) letter score of 19 to 48 ETDRS letters (Snellen equivalent of 20/125–20/400). Geographic atrophy was not required in the nonstudy eye. The key exclusion criteria included: GA in the study eye due to causes other than AMD; uncontrolled glaucoma; current vitreous hemorrhage; prior laser photocoagulation or other ophthalmic treatments, including photodynamic therapy, external beam radiation therapy, or transpupillary thermotherapy; history of choroidal neovascularization in either eye; retinal pigment epithelium tear that involves the macula in either eye; and moderate or severe nonproliferative diabetic retinopathy in either eye.

Study Design and Treatment

The study consisted of the following 2 stages: a single ascending dose stage with 4 planned IVT dose levels (cohorts A–D at 1 mg, 3 mg, 10 mg, and 20 mg, respectively) to evaluate single-dose safety and tolerability of RO7303359, and an expansion cohort stage at the MTD or maximum tested dose to further characterize safety, PK, and potential PD biomarkers of target engagement and modulation of downstream IL-33 pathway activity (Fig 1). Per protocol, an optional cohort could be opened to further characterize study objectives.

The study was divided into the following 3 periods: a screening period lasting up to 28 days, a treatment period consisting of a single day (day 1) during which RO7303359 was administered, and a 12-week follow-up period. The study allowed graded exposure to RO7303359 with escalation to the next higher-dose cohort based on safety and tolerability data for ≥3 patients from the preceding dose cohort(s) after 14 days of follow-up had elapsed for the last patient in the cohort.

Maximum tolerated dose was defined as the dose level below the one that met the dose-limiting toxicity criteria. If the dose-limiting toxicity criteria were met at the highest dose level administered in this study, this dose level was declared the maximum tested dose. A dose-limiting adverse event (AE) was defined as uveitis or vitritis exceeding the inflammatory reaction expected after an IVT injection on the day after treatment (day 2) or as a change of 2 units on standard grading scales as assessed on days 3 to 14 after study drug injection; or loss of visual acuity ≥15 letters that occurred during the dose-limiting AE assessment window (14 days after study drug injection), unless attributed by the investigator to another clearly identifiable cause.

Safety and Tolerability Assessments

Safety was assessed based on frequency, severity, and timing of ocular and systemic AEs, graded in severity according to the World Health Organization toxicity grading scale. It also included assessment of changes in BCVA, intraocular pressure, slit lamp findings, indirect ophthalmoscopy, spectral-domain OCT, vital signs, and physical findings.

Ocular Assessments

At each study visit, the following ocular assessments were performed: BCVA as determined by ETDRS chart, intraocular pressure measurement, slit lamp examination, and dilated binocular indirect ophthalmoscopy. Postinjection intraocular pressure was measured in the study eye 30 to 50 minutes after injection at the day 1 study treatment visit. In addition, the following ocular images were obtained from both eyes: fluorescein angiography (at screening, week 12, and at termination), digital color fundus photography (at screening, week 12), and spectral-domain OCT (at screening; study days 2, 3, and 8; weeks 2, 4, 8, and 12; or early termination visit).

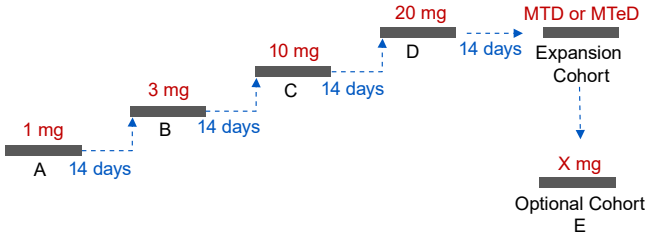


Figure 1. Study design. (**A**–**E**) Indicate the dose level cohorts. MTD = maximum tolerated dose; MTeD = maximum tested dose.

PKs and PDs

The PK properties of RO7303359 were characterized after single IVT doses based on serum and aqueous humor concentrations at specified time points. Serum samples were collected on study days 1, 2, 3, and 8, then weeks 2, 4, 8, and 12; aqueous humor samples were collected on study days 1 (for PD evaluation only) and 8, and weeks 4, 8, and 12.

RO7303359 serum and aqueous humor concentrations were measured using validated immunoaffinity liquid chromatography with tandem mass spectrometry methods. Streptavidin magnetic beads (Cytiva) coupled to biotinylated anticomplementarity determining region antibody to RO7303359 (Genentech) were used for enrichment. Bound proteins were subjected to on-bead proteolysis with trypsin. Unique peptide fragments used as analytical surrogates for RO7303359 were used for quantitative analysis by liquid chromatography with tandem mass spectrometry. The minimum quantifiable concentration was 0.40 ng/mL in serum and 1.0 ng/mL in aqueous humor.

Estimates of total exposure (area under the concentration-time curve), maximum observed concentration, time of maximum observed concentration, and terminal half-life were based on noncompartmental analysis.

The PD objectives for this study included the evaluation of aqueous humor biomarker levels at specified time points predictive of response to RO7303359 and evidence of activity. Aqueous humor samples were collected and evaluated on study days 1 and 8, and weeks 4, 8, and 12. Aqueous humor biomarkers selected for evaluation included CC motif chemokine ligand 2 (CCL2; also known as monocyte chemoattractant protein-1), CXC motif chemokine ligand 10 (CXCL10), intercellular adhesion molecule 1 (ICAM1), and IL-6. These biomarkers were selected based on preclinical evidence that they represent downstream mediators of IL-33/ST2 pathway activity,[9] and clinical evidence from a cross-sectional study in which CCL2, CXCL10, ICAM1, and IL-6 were elevated in the aqueous humor of GA secondary to AMD subjects compared with non-AMD control subjects, with median deltas of 16%, 21%, 34%, and 30%, respectively.[13] Due to sample size limitations, a prespecified threshold for biomarker level reduction was not established. However, the use of multiple biomarkers and the expectation that consistent reductions across them would be necessary to demonstrate a PD effect served to mitigate this limitation. Evaluation for activity using clinical measures (e.g., change in GA lesion area on fundus autofluorescence imaging) was not included in the study design because it is not possible to assess progression over the 12 week follow-up period.

Biomarker levels were measured in aqueous humor by a qualified multiplex immunoassay (Ella 4-plex from ProteinSimple) at baseline (day 1) and at multiple time points (day 8, week 4, week 8, and week 12) after RO7303359 treatment. Measurements were done for 3 patients each in the 1-mg, 3-mg, and 10-mg dose cohorts, and for 14 biomarker-evaluable patients in the 20-mg MTD cohort. The lower limit of quantification for these biomarkers is: CCL2 (6.08 pg/mL), CXCL10 (2.4 pg/mL), IL-6 (1.12 pg/mL), and ICAM1 (16.4 pg/mL). Levels of the reduced form of IL-33 were also evaluated in aqueous humor in the same patients at the same time points using a qualified immuno-polymerase chain reaction assay. The lower limit of quantification of the assay is 0.97 pg/mL.

Immunogenicity

For immunogenicity assessment of the presence of antidrug antibodies (ADAs) to RO7303359, serum samples were collected on study day 1 (baseline) and 8; weeks 2, 4, and 12; and any unscheduled visits. A validated semihomogeneous, bridging enzyme-linked immunosorbent assay was used to measure the levels of serum ADAs to RO7303359. Antibody titer values were calculated by log transformation of the dilution factor at which the titrated sample crossed the assay cut point. The minimum reportable titer was 1.3.

Statistical Analyses

The sample size was selected without performing a power calculation to provide descriptive information on safety, tolerability, PK, and PD information after RO7303359 administration. The initial sample size of 3 to 6 patients per cohort was chosen to obtain reasonable evidence of safety and tolerability without exposing undue numbers of patients to RO7303359 during the dose-escalation phase. Up to 15 patients in the highest dose (MTD or maximum tested dose) cohort were to be enrolled to obtain $\geq$10 evaluable patients to fulfill further safety assessment and PD activity assessment of RO7303359. The safety analyses included all patients who received a dose of study drug, with patients grouped according to treatment received. Given the nature of the study, all statistical summaries were descriptive in nature; no formal statistical hypothesis testing was performed.

Data Sharing

For eligible studies, qualified researchers may request access to individual patient-level clinical data through a data request platform. At the time of writing this request platform is Vivli: https://vivli.org/ourmember/roche/. For up to date details on Roche's Global Policy on the Sharing of Clinical Information and how to request access to related clinical study documents, see here: https://www.roche.com/innovation/process/clinical-trials/data-sharing. Anonymized records for individual patients across >1 data source external to Roche cannot, and should not, be linked due to a potential increase in risk of patient reidentification.

Results

Patient Characteristics and Disposition

A total of 37 participants were enrolled in the study, all of whom received a single injection of study drug. We enrolled

Table 1. Patient Demographics and Characteristics

Characteristic	1-mg RO7303359 (n = 3)	3-mg RO7303359 (n = 3)	10-mg RO7303359 (n = 3)	20-mg RO7303359 (n = 3)	20-mg Expansion RO7303359 (n = 12)	10-mg Optional RO7303359 (n = 13)	All Participants (N = 37)
Age in years, mean (range)	73.7 (71–76)	74.7 (65–89)	87.1 (79–83)	76.3 (67–87)	83.3 (59–96)	77.8 (63–91)	79.2 (59–96)
Sex, n (%)							
Male	0	2 (66.7)	0	1 (33.3)	5 (41.7)	6 (46.2)	14 (37.8)
Female	3 (100)	1 (33.3)	3 (100)	2 (66.7)	7 (58.3)	7 (53.8)	23 (62.2)
Ethnicity, n (%)							
Not Hispanic or Latino	3 (100)	3 (100)	3 (100)	3 (100)	12 (100)	12 (92.3)	36 (97.3)
Not stated	0	0	0	0	0	1 (7.7)	1 (2.7)
GA area (mm^2) by FP, total area (SD)	18.10 (2.49)	24.54 (4.19)	29.57 (6.33)	22.97 (10.95)	16.47 (14.62)	11.17 (8.08)	16.98 (11.43)
No. atrophic spots by FP, mean (SD)	1.00 (0.00)	1.00 (0.00)	1.33 (0.58)	1.00 (0.00)	1.08 (0.29)	1.38 (0.87)	1.19 (0.57)
ETDRS visual acuity, score (SD)	36.0 (8.5)	34.7 (6.1)	24.3 (3.8)	27.3 (5.1)	31.1 (10.5)	31.5 (9.2)	31.1 (8.8)
ETDRS BCVA categories, approximate Snellen equivalent (%)							
20/200 or worse	2 (66.7)	2 (66.7)	3 (100)	3 (100)	9 (75.0)	10 (76.9)	29 (78.4)
Better than 20/200	1 (33.3)	1 (33.3)	0	0	3 (25.0)	3 (23.1)	8 (21.6)
Intraocular pressure (mmHg), mean (range)	14.0 (10–17)	14.3 (12–17)	15.0 (11–20)	17.7 (16–19)	12.8 (10–18)	15.3 (10–21)	14.5 (10–21)

BCVA = best-corrected visual acuity; FP = fundus photography; GA = geographic atrophy; SD = standard deviation.
Baseline ophthalmic characteristics are for the study eye.

1 optional cohort, terminated early after 13 of the target 15 patients were enrolled and completed the study. Biomarker analysis was not done on the optional cohort. Of these participants, 1 (2.7%) discontinued from the optional cohort due to an AE of back pain, which was deemed unrelated to study drug. Baseline and demographic characteristics are summarized in Table 1. Overall, the majority of patients were female (62.2%) and White (91.9%) with the median age of 81 years (range 59–96 years). Ocular baseline characteristics are shown in Table 1. At baseline, the mean (standard deviation) study eye BCVA letter score was 31.1 (8.8) letters. The mean (standard deviation) total GA area, as determined by fundus photography, was 16.98 (11.43) mm^2 in the study eye of enrolled patients.

Safety

Single IVT injections of RO7303359 showed acceptable safety in patients with GA secondary to AMD. No dose-limiting AEs were reported in this study. Overall, 22 of 37 participants (59.5%) had ≥1 AE. All reported AEs were mild to moderate in severity, with no greater than or equal to grade 3 AEs reported. Nonserious grade 1 photophobia in both eyes, in a participant receiving 20-mg ITV RO7303359, was the only AE assessed as related to study treatment.

Ocular AEs

Nineteen of the 37 enrolled participants (51.4%) reported ≥1 ocular AE in the study eye (Table 2). Two (5.4%) reported ≥1 ocular AE in the fellow eye. The only AE reported in >1 study participant was conjunctival hemorrhage or hyperemia (16 of 37 [43%]). There was 1 nonserious grade 1 iritis reported as related to the study procedure, on day 2, resolving on day 3 without treatment, in the 10 mg optional cohort. Examination by slit lamp revealed anterior chamber cell 0.5+ and anterior

Table 2. Incidence of Ocular Adverse Events

Ocular adverse event	1-mg RO7303359 (n = 3)	3-mg RO7303359 (n = 3)	10-mg RO7303359 (n = 3)	20-mg RO7303359 (n = 3)	20-mg Expansion RO7303359 (n = 12)	10-mg Optional RO7303359 (n = 13)	All Participants (N = 37)
Study eye							
Conjunctival hemorrhage, n (%)	1 (33.3)	2 (66.7)	2 (66.7)	3 (100)	3 (25.0)	3 (23.1)	14 (37.8)
Conjunctival hyperemia, n (%)	1 (33.3)	0	1 (33.3)	0	0	0	2 (5.4)
Conjunctival edema, n (%)	1 (33.3)	0	0	0	0	0	1 (2.7)
Iritis, n (%)	0	0	0	0	0	1 (7.7)	1 (2.7)
Increased lacrimation, n (%)	0	0	0	0	1 (8.3)	0	1 (2.7)
Photophobia, n (%)	0	0	0	0	1 (8.3)	0	1 (2.7)
Nonstudy eye							
Anterior chamber cell	0	0	0	0	0	1 (7.7)	1 (2.7)
Photophobia	0	0	0	0	1 (8.3)	0	1 (2.7)

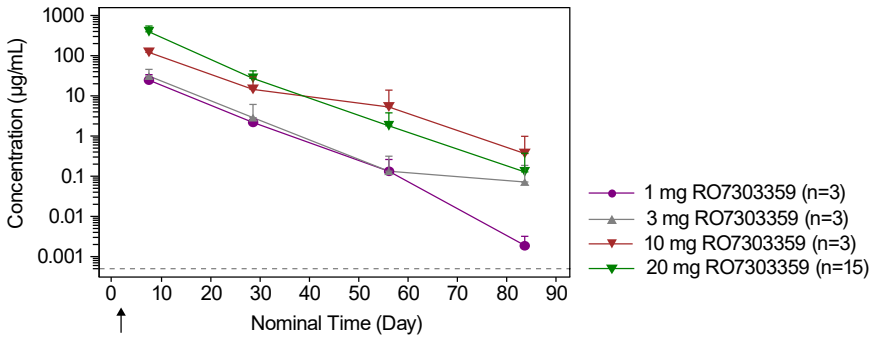


Figure 2. RO7303359 aqueous humor pharmacokinetics. The mean (+SD) concentration-time profiles are shown for each dose cohort. Arrow represents dose administration. Below limit of quantification results on postdose samples are set to 0.0005 µg/mL, that is, half of the minimum quantifiable concentration value (0.001 µg/mL). SD = standard deviation.

chamber flare 1+. Apart from this event, there were no other instances of ocular inflammation reported in this study. There were no ocular serious AEs reported in study or nonstudy eyes. All ocular AEs resolved before study exit.

Nonocular AEs

Six (16.2%) of all enrolled participants had ≥1 systemic AE (Table S1, available at www.ophthalmologyscience.org). None of the AEs were reported in >1 study participant. All nonocular AEs reported were nonserious, mild or moderate in severity, and no deaths were reported.

PK and Immunogenicity

Serum concentration−time profiles for RO7303359 in participants after IVT injection are shown in Figure S1 (available at www.ophthalmologyscience.org). After IVT injection, the mean terminal half-life, based on serum PK data, ranged from 5.21 to 6.51 days across cohorts, with a mean coefficient of variation (CV%) of 13.9% to 73.1%. The serum maximum observed concentration and area under the concentration-time curve generally demonstrated dose proportionality, with CV% ranging from 19.2% to 45.3%. Time of maximum observed concentration was observed approximately 1 day after IVT injection.

The PK profile of RO7303359 in aqueous humor was generally consistent with the serum PK profile. Aqueous humor concentration-time profiles after IVT injection are shown in Figure 2. After IVT injection, the mean terminal half-life in aqueous humor ranged from 5.39 to 8.59 days across cohorts, with a mean CV% of 12.9% to 78.7%. The aqueous humor maximum observed concentration and area under the concentration-time curve had moderate variability, with CV% ranging from 13.2% to 68.2%. The time of maximum observed concentration was observed approximately 7 days (at the first postinjection sampling point) after IVT injection.

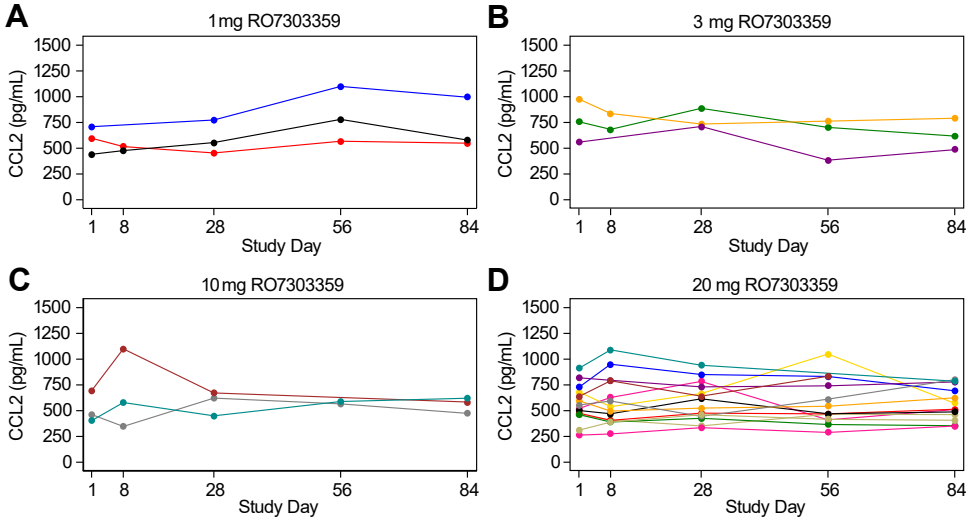


Figure 3. CCL2 levels in aqueous humor after RO7303359 administration. Each colored line represents 1 patient. **A,** 1 mg, **(B)** 3 mg, **(C)** 10 mg, and **(D)** combined with MTD expansion cohort (20 mg). CCL2 = CC motif chemokine ligand 2; MTD = maximum tolerated dose.

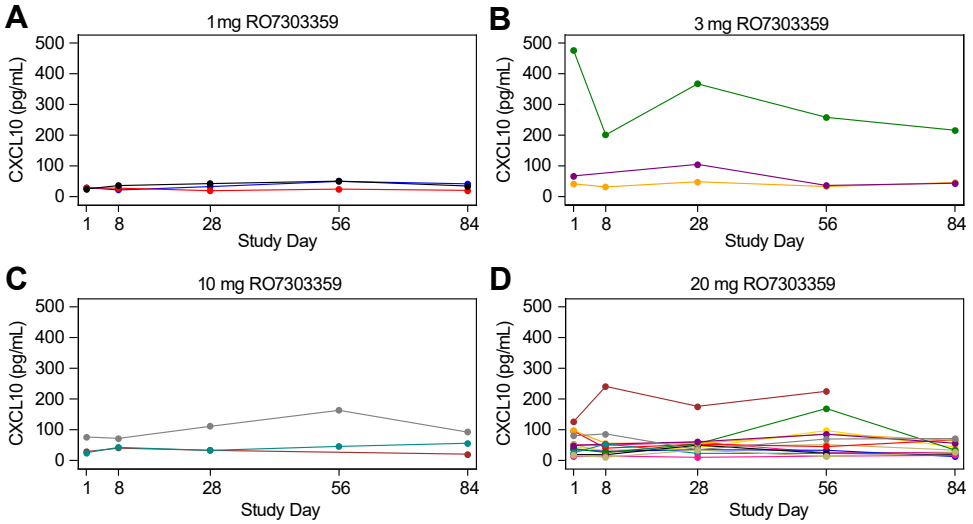


Figure 4. CXCL10 levels in aqueous humor after RO7303359 administration. Each colored line represents 1 patient. **A,** 1 mg, **(B)** 3 mg, **(C)** 10 mg, and **(D)** combined with MTD expansion cohort (20 mg). CXCL10 = CXC motif chemokine ligand 10; MTD = maximum tolerated dose.

The baseline prevalence of anti-RO7303359 ADA was 0.0% (0 of 26 ADA-evaluable patients). The incidence of treatment-emergent ADA was 0.0% (0 of 27 ADA-evaluable patients).

PD

Aqueous humor levels of CCL2 (also known as monocyte chemoattractant protein-1), CXCL10, IL-6, and ICAM1 were assessed as PD biomarkers of RO7303359 inhibition of IL-33/ST2 pathway activity. The absolute values at each visit versus time for all biomarker evaluable patients and across cohorts A (1 mg), B (3 mg), C (10 mg), and D combined with MTD expansion cohort (20 mg) are summarized in Figures 3 to 6. There was no discernable PD effect (reduction from baseline values) observed with any of the 4 biomarkers in any of the dosing cohorts. The percent changes from baseline for the 4 evaluated aqueous humor biomarkers versus time for the 20-mg dose MTD cohort with 14 biomarker evaluable patients are summarized in Figure 7. This analysis also shows that there is no discernable PD effect. Outlier IL-6 elevations, observed in a small patient subset, skewed the mean but were not associated with clinical findings or adverse effects. Systemic IL-6 was not measured. The observations in the 20 mg MTD make it unlikely that PD effects would be observed in the 10-mg dose optional cohort, and hence samples from the patients in that cohort were not analyzed and instead saved for future research on GA.

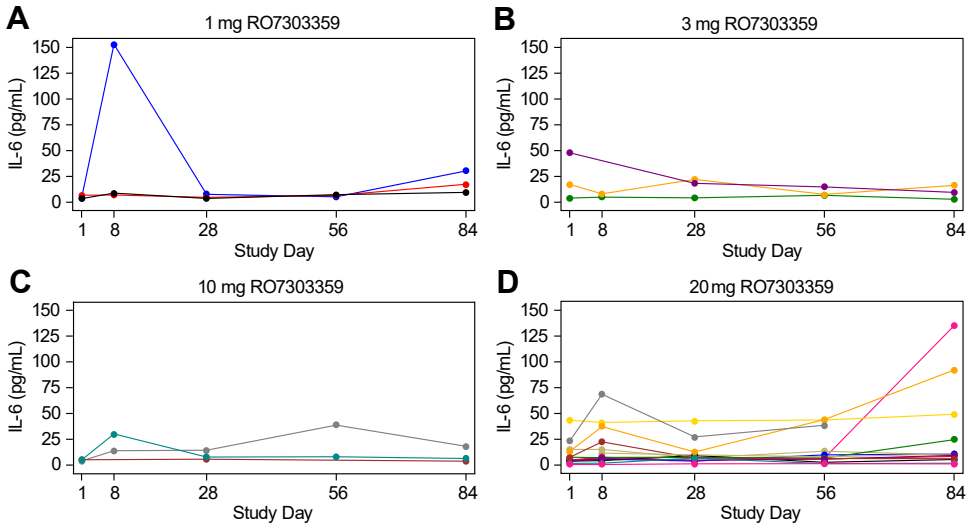


Figure 5. Interleukin 6 levels in aqueous humor after RO7303359 administration. Each colored line represents 1 patient. **A,** 1 mg, **(B)** 3 mg, **(C)** 10 mg, and **(D)** combined with MTD expansion cohort (20 mg). IL = interleukin; MTD = maximum tolerated dose.

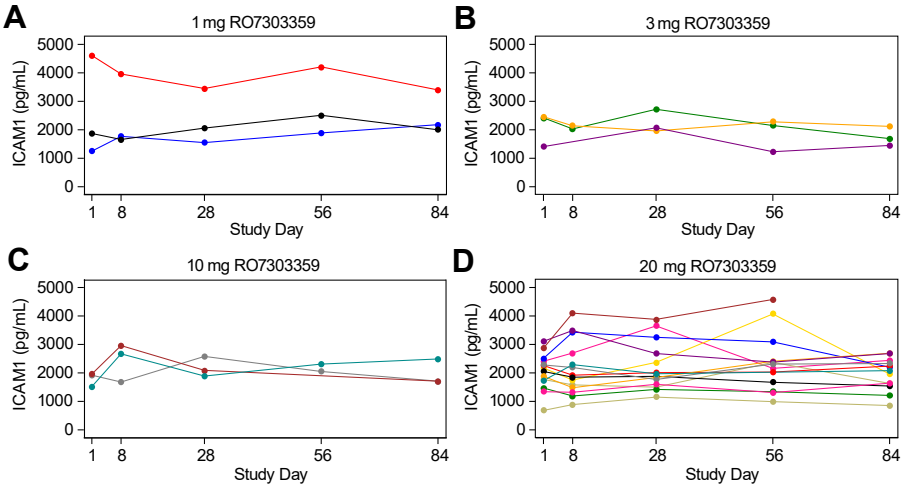


Figure 6. ICAM1 levels in aqueous humor after RO7303359 administration. Each colored line represents 1 patient. **A,** 1 mg, **(B)** 3 mg, **(C)** 10 mg, and **(D)** combined with MTD expansion cohort (20 mg). ICAM1 = intercellular adhesion molecule 1; MTD = maximum tolerated dose.

Levels of reduced form of IL-33 were below the lower limit of quantification (0.97 pg/mL) at all time points including baseline (predose). This could be due to any of the following: low concentration of the target and/or the reduced form of IL-33 is not stable for long enough to be detected in aqueous humor even in the presence of RO7303359. Hence, we cannot draw conclusions about target engagement in this study.

Discussion

Herein, we report the results of a multicenter, single-arm, single-dose, dose-escalation, first-in-human study to evaluate IVT administration of an anti−IL-33 Fab (RO7303359) in patients with GA secondary to AMD. Targeting inhibition of the IL-33/ST2 pathway to mediate inflammatory processes in the retinal pigment epithelium represents a novel strategy for treatment of AMD, with rationale based on data from immunohistochemical analyses of human eyes and nonclinical models of phototoxic retinal degeneration.[9] To our knowledge, this is the first report of a clinical trial to evaluate inhibition of the IL-33/ST2 pathway inside the eye.

RO7303359 demonstrated an acceptable safety profile for doses up to 20 mg, with the caveat that this was a small, single-dose study. A single case of mild iritis noted on the day after RO7303359 administration seems likely to be related to the IVT injection and aqueous humor sampling procedures, given inflammation resolution within 1 day without any intervention. The absence of treatment-emergent ADA among evaluable patients was not surprising for a single-dose study.

Clinical measures of activity for therapeutics intended to slow or arrest progression of GA secondary to AMD do not exist for studies of short duration. Evidence for PD effect was sought by evaluating for a decrease in levels of CCL2,

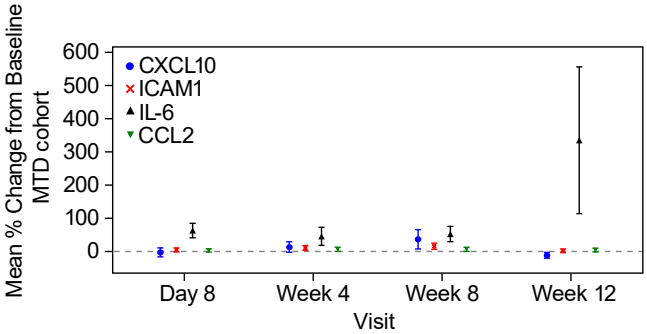


Figure 7. Aqueous humor percent change from baseline for aqueous humor biomarker levels by visits in MTD cohort. The mean (±SE) percent change from baseline concentration for each biomarker in the MTD cohort is shown. CCL2 = CC motif chemokine ligand 2; CXCL10 = CXC motif chemokine ligand 10; ICAM1 = intercellular adhesion molecule 1; IL = interleukin; MTD = maximum tolerated dose; SE = standard error.

CXCL10, IL-6, and ICAM1 in aqueous humor after RO7303359 administration in a subset of patients, including those given the maximal (20 mg) dose. These 4 analytes were prespecified based on in vitro and rodent studies suggesting their role in downstream signaling of the IL-33/ST2 pathway[9,13] and based on cross-sectional data suggesting elevated levels in aqueous humor of eyes with GA secondary to AMD relative to controls.[13] Blocking the IL-33/ST2 pathway reduced retinal CCL2 and IL-6 in mice.[9] Interleukin 33 stimulation increased all 4 selected biomarkers in cell culture experiments (Genentech data on file), and they were all elevated in aqueous humor of patients with GA.[13] No reduction in levels of CCL2, CXCL10, IL-6, and ICAM1 was observed after administration of RO7303359. The absence of changes in biomarker levels, despite preclinical evidence suggesting their relevance to IL-33/ST2 pathway activity, weakens the hypothesis of this pathway's significant role in GA progression. However, limitations in the preclinical model, including the acute phototoxicity's translatability, as well as aqueous humor's representativeness of the pathobiology of GA lesions, should be considered.

The measured half-life of RO7303359 was between approximately 5 and 9 days based on PK analysis of serum and aqueous humor samples, within the typical range for a Fab administered intravitreally. The PK profile, in combination with the absence of treatment-emergent ADA, suggests expected exposure inside the eye, but intraocular target engagement could not be confirmed because measurement of reduced (active) IL-33 was undetectable (below the lower limit of quantification of the assay) for all tested aqueous humor samples (cohorts A, B, C, and combined D and E [MTD]). The reduced form of IL-33 is known to be rapidly oxidized and inherently unstable, offering a likely explanation for the inability to detect it in aqueous humor samples.

The development of RO7303359 has been discontinued by the sponsor based on the results of the PD analysis, not the safety profile. Despite the limitations of the study, including its small sample size, single-dose design, and short duration, and the caveat that biomarker assessments in aqueous humor might not reflect IL-33 activity in the retina, the absence of target engagement biomarker evidence did not support a more robust assessment of efficacy in late stage development.

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Abbreviations and Acronyms:
AE = adverse event; **ADA** = antidrug antibody; **AMD** = age-related macular degeneration; **BCVA** = best-corrected visual acuity; **CCL2** = CC motif chemokine ligand 2; **CV%** = coefficient of variation; **CXCL10** = CXC motif chemokine ligand 10; **Fab** = fragment antigen-binding region; **GA** = geographic atrophy; **ICAM1** = intercellular adhesion molecule 1; **IL** = interleukin; **IVT** = intravitreal; **MTD** = maximum tolerated dose; **PD** = pharmacodynamic; **PK** = pharmacokinetic; **ST2** = serum stimulation-2.

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